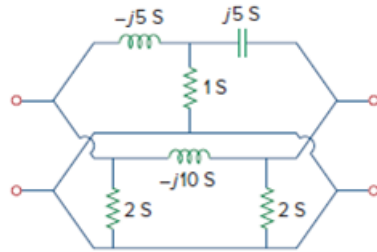


Problem

Obtain the y parameters for the network in Fig. 19.45.

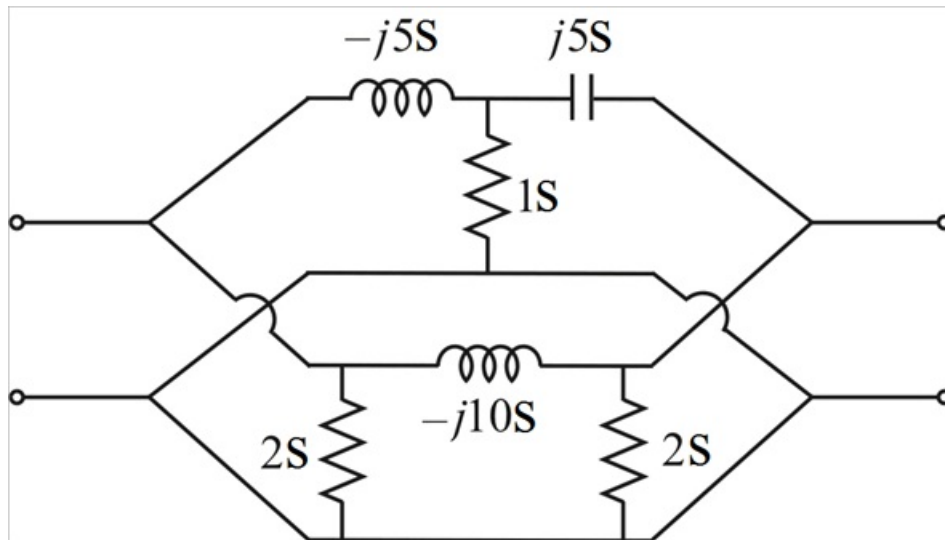
Figure 19.45



Step-by-step solution

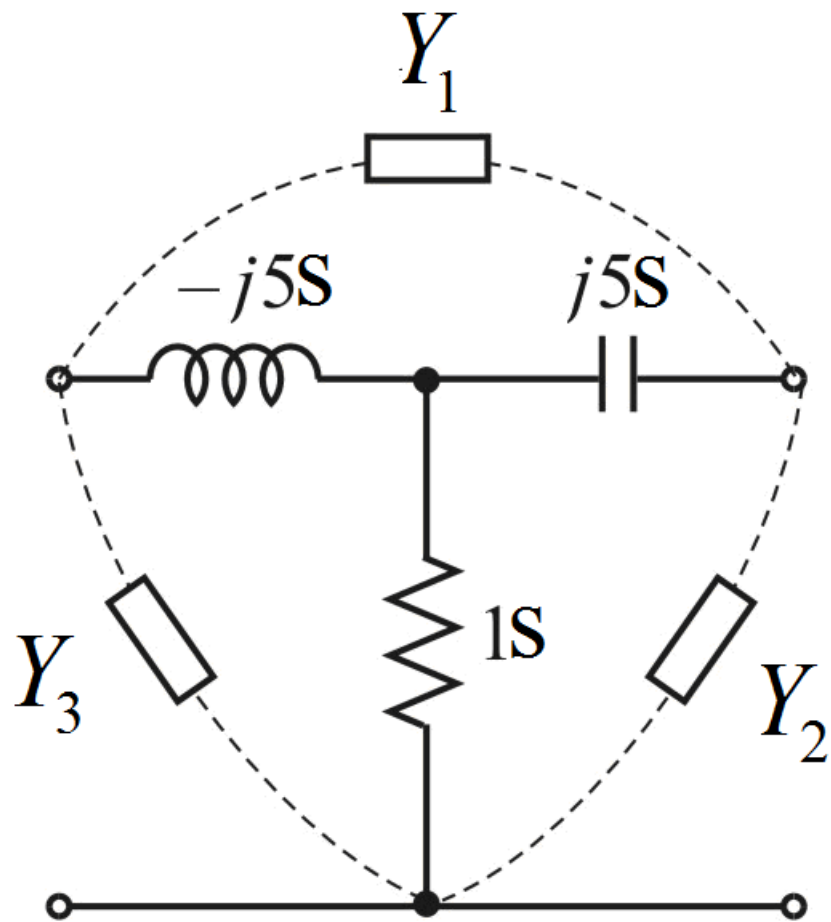
Step 1 of 6

Given circuit



Step 2 of 6

Consider upper two – port network as N_a



Step 3 of 6

Convert T – network into Π network,

By which

$$R_1 = \frac{\left(\frac{1}{1}\right)\left(\frac{1}{-5j}\right) + \left(\frac{1}{1}\right)\left(\frac{1}{j5}\right) + \left(\frac{1}{5j}\right)\left(\frac{1}{-j5}\right)}{\frac{1}{1}}$$

$$R_1 = \frac{0.04}{1}$$

$$\therefore Y_1 = 25S$$

$$R_2 = \frac{\left(\frac{1}{1}\right)\left(\frac{1}{-5j}\right) + \left(\frac{1}{1}\right)\left(\frac{1}{j5}\right) + \left(\frac{1}{5j}\right)\left(\frac{1}{-j5}\right)}{\frac{1}{-5j}}$$

$$R_2 = \frac{0.04}{\frac{1}{-5j}}$$

$$R_2 = -j5(0.04)$$

$$R_2 = -j0.2$$

$$Y_2 = j5$$

$$R_3 = \frac{\left(\frac{1}{1}\right)\left(\frac{1}{-5j}\right) + \left(\frac{1}{1}\right)\left(\frac{1}{j5}\right) + \left(\frac{1}{5j}\right)\left(\frac{1}{-j5}\right)}{\frac{1}{5j}}$$

$$R_3 = \frac{0.04}{\frac{1}{5j}}$$

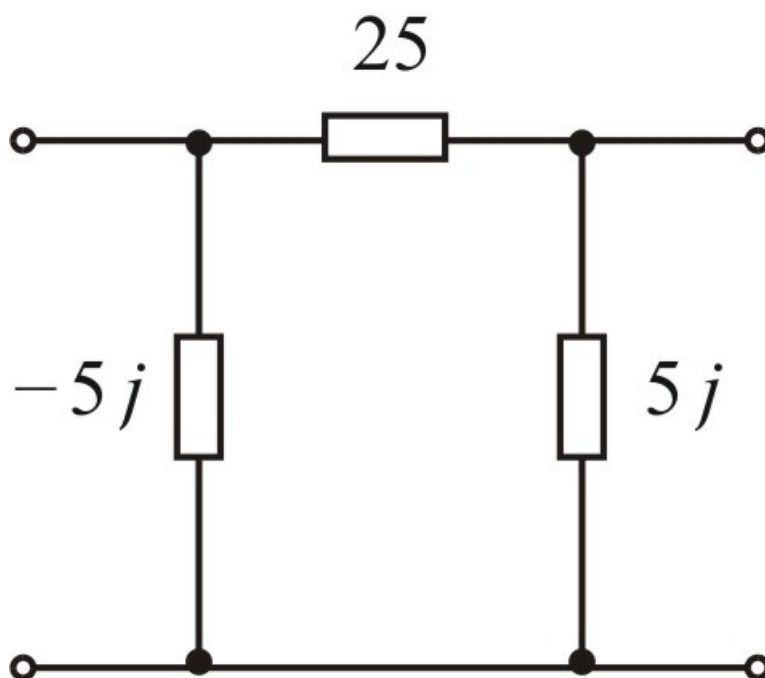
$$R_3 = j5(0.04)$$

$$R_3 = j0.2$$

$$Y_3 = -j5$$

Step 4 of 6

Equivalent Π network is



Step 5 of 6

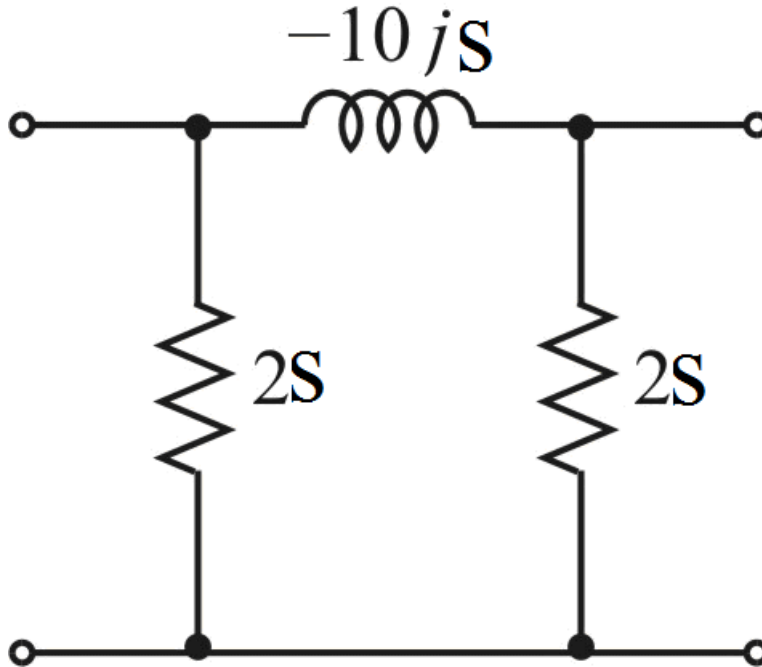
$$y_{12a} = y_{21a} = -25,$$

$$y_{11a} = 25 - 5j$$

$$y_{22a} = 25 + 5j$$

$$[y_a] = \begin{bmatrix} 25 - 5j & -25 \\ -25 & 25 + 5j \end{bmatrix} \text{S}$$

Consider the lower network as N_b



Step 6 of 6

$$y_{12b} = y_{21b} = 10j$$

$$y_{11b} = 2 - 10j$$

$$y_{22b} = 2 - 10j$$

$$[y_b] = \begin{bmatrix} 2 - 10j & 10j \\ 10j & 2 - 10j \end{bmatrix}$$

Thus, the two networks are connected in parallel. Therefore, the overall y parameters are

$$[y] = [y_a] + [y_b]$$

$$[y] = \begin{bmatrix} 25 - 5j & -25 \\ -25 & 25 + 5j \end{bmatrix} + \begin{bmatrix} 2 - 10j & 10j \\ 10j & 2 - 10j \end{bmatrix}$$

$$\therefore [y] = \begin{bmatrix} 27 - 15j & -25 + 10j \\ -25 + 10j & 27 - 5j \end{bmatrix} \text{S}$$